Digital Fundamental : DPP of Lec 12

Computer Science

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Q1 What is the sum output of a half adder when both inputs are 1?

(A) 0 (B) 1

(C) 2 (D) undefined

Q2 Which of the following statements is /are true for Combinational circuits?

I: Combinational circuits are defined as time-independent circuits.

II: No memory elements are required as they do not store past inputs.

III: Generally faster since they don't rely on previous states.

IV: Examples of combinational circuits are counters, shift registers, and memory units.

(A) Only I

(B) Only II

(C) Only I, II, and III

(D) Only IV

Q3 What are the correct Boolean expressions for the **sum** and **carry** outputs of a half-adder?

(A) Sum = A+B, Carry = A B

(B) Sum = A B, Carry = A+B

(C) Sum = A B, Carry = A B

(D) More than one of the above

Q4 How many select lines will 16 to 1 multiplexer have?

(A) 4 (B) 3

(C) 5 (D) 1

Q5 **Match the following combinational circuit components in Column A with Column B:**

Column A	Column B
1. **Multiplexer**	a) Selects one input from many inputs and forwards it to a single output
2. **Encoder**	b) Converts binary data from n input lines into 2^n output lines
3. **Decoder**	c) Converts 2^n input lines to n output lines
4. **Full-Adder**	d) Performs addition of two bits and a carry bit, producing sum and carry output

(A) 1 – a, 2 – d, 3 – b, 4 – c

(B) 1 – a, 2 – c, 3 – b, 4 – d

(C) 1 – c, 2 – a, 3 – b, 4 – d

(D) 1 – b, 2 – c, 3 – a, 4 – d

Q6 Which of the following are an example of combinational circuit?

(A) Multiplexer

(B) Encoder

(C) Full adder

(D) All of the above

Q7 A full adder can be implemented with _____ and _____ gates.
(A) Half adders and OR gate
(B) Half adders and NOT gate
(C) Multiplexer and OR gate
(D) None of the above

Q8 How many full adders are required to add two 4-bit binary numbers?
(A) 2 (B) 3
(C) 4 (D) 5

Q9 Which of the following statements about full adders is true?
(A) They can only add single bits.
(B) They can handle carry propagation.
(C) They do not have a carry output.
(D) They can only add numbers without carry input.

Q10 If a Full adder has two inputs A, B, C_{in} and two outputs sum S and carry C_{out} then the expression for sum S and carry C_{out} are _____.
(A) CARRY-OUT = AB + BC_{in} + AC_{in} SUM = (A B) C_{in}
(B) CARRY-OUT = AB + BC_{in} + C_{in} SUM = (A B) C_{in}
(C) CARRY-OUT = AB + BC_{in} + AC_{in} SUM = (A B) C_{out}
(D) None of the above

Answer Key

Q1	A
Q2	C
Q3	C
Q4	A
Q5	B
Q6	D
Q7	A
Q8	C
Q9	B
Q10	A

Hints & Solutions

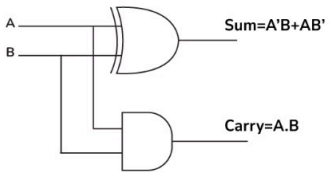
Note: scan the QR code to watch video solution

Q1 Text Solution:

Logical Expression of Half Adder

Sum (S) = A B

Carry (C) = A . B



Truth Table

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Q2 Text Solution:

Statement I: Combinational circuits are defined as time-independent circuits that do not depend upon previous inputs to generate any output.

Statement II: The output depends only upon the present input and there is no need for feedback for input and output, so a memory element is not required.

Statement III: Combinational circuits are faster than sequential circuits since they don't rely on previous states.

Statement IV: Counters, shift registers, and memory units are examples of Sequential circuits. so this statement is wrong for combinational circuits.

Q3 Text Solution:

A **half-adder** adds two single-bit binary numbers, **A** and **B**, and outputs a **sum** and a **carry**.

Boolean Expressions:

The sum is calculated using an XOR (exclusive OR) operation between A and B.

The **carry** is calculated using an **AND** operation between **A** and **B**.

Correct Boolean Expressions:

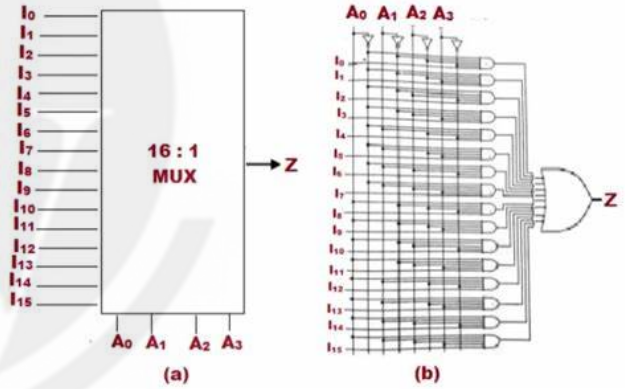
Sum = A B (XOR operation)

Carry = A · B (AND operation)

Option C is correct.

Q4 Text Solution:

In 16 t0 1 MUX, four select lines will be required to select 16 (2^4) inputs.



Q5 Text Solution:

The correct matches are:

1. Multiplexer - a) Selects one input from many inputs and forwards it to a single output.

2. Encoder -c) Converts 2^n input lines to *n* output lines.

3. Decoder - b) Converts binary data from *n* input lines into 2^n output lines.

4. Full-Adder - d) Performs the addition of two bits and a carry bit, producing sum and carry output.

Q6 Text Solution:

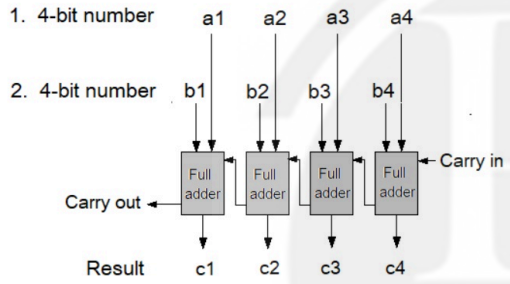
All the options listed—Multiplexer, Encoder, and Full Adder—are examples of combinational circuits. Combinational circuits are those where the output is a pure function of the present input values, with no memory or storage elements involved.

Q7 Text Solution:

A full adder can be implemented using two half adders and one OR gate. The half adders handle the addition of the bits, and the OR gate combines the carry outputs.

Q8 Text Solution:

For summing two 4-bit numbers we need 4 full adders.



Q9 Text Solution:

Full Adder: A full adder is a digital circuit that adds three binary digits: two significant bits and a carry bit from a previous addition. It has three inputs (A, B, and Carry-in) and produces two outputs: the Sum and the Carry-out.

Carry-in and Carry-out:

Carry-in: This is the carry from the addition of less significant bits. It allows the full adder to add values that exceed the binary limit (1 + 1 = 10 in binary, where 0 is the sum and 1 is carried to the next higher bit).

Carry-out: This is the output carry that results from the addition, which can be sent to the next more significant bit in a multi-bit addition.

Q10 Text Solution:

Logical Expression of Full Adder

CARRY-OUT = $AB + BC_{in} + AC_{in}$

SUM = (A B) C_{in}

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